MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 6SL7WGT AND 12SL7GT

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin triode, high-Mu

Ⓒ

Outline		
6SL7WGT	---	See note 1
12SL7GT	---	9-11 or 9-41 (EIA)
Base		
6SL7WGT	---	B8-6 or B8-46 (Low-loss phenolic)
12SL7GT	---	B8-6 or B8-46 (Phenolic)
Envelope	---	T9
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2g	2a	2k	1g	1a	1k	h	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Pp/a	Ⓒ Ehk	Ⓒ Alt
Unit:	V	Vdc	Vdc	W	v	ft
Maximum:						
6SL7WGT	6.9	275	---	1.1	100	Note 2
12SL7GT	13.9	275	---	1.1	100	Note 2
Minimum:						
6SL7WGT	5.7	---	---	---	---	---
12SL7GT	11.3	---	---	---	---	---
TEST CONDITIONS:						
6SL7WGT	6.3	250	-2	---	---	---
12SL7GT	12.6	250	-2	---	---	---

GENERAL:

Qualifications - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
1031	Variable-frequency vibration 6SL7WGT	Rp = 2,000 ohms (see note 4)	---	---	Ep	---	200	mVac
1031	Vibration fatigue 6SL7WGT		---	---	---	---	---	---
---	Vibration-fatigue test end points 6SL7WGT	Low-frequency vibration	---	---	Ep	---	200	mVac
		Heater-cathode leakage	---	---	Ihk	---	45	µAdc
		Transconductance	---	---	Sm	1,040	---	µmhos
---	Mechanical resonance 6SL7WGT	See note 3	---	---	---	---	---	---
	Quality conformance inspection, part 1							
1266	Total grid current	See notes 4 and 5	0.65	II	Ic	0	-1.0	µAdc
1321	AC Amplification	Ebb = 100 Vdc; Ecc = 0; Esig = 0.2 Vac (see note 6)	0.65	II	Ep	7.5	---	Vac
Ⓒ 1246	Audio frequency noise	Rp = 0.2 Meg; Esig = 70 mVac (see note 4)	0.65	II	---	---	17	vu
1231	Emission	Eb = Ec = 30 Vdc (see notes 5 and 6)	0.65	II	Is	40	---	mAdc
Ⓒ 1201	Short and discontinuity detection		0.4	II	---	---	---	---
1256	Electrode current (1) (anode)	See note 6	0.65	II	Ib	1.4	3.2	mAdc
	Quality conformance inspection, part 2							
1031	Low-frequency vibration 6SL7WGT	Rp = 2,000 ohms (see note 4)	---	---	Ep	---	200	mVac
	12SL7GT	Rp = 2,000 ohms (see note 4)	---	---	Ep	---	400	mVac
Ⓒ 1041	Shock 6SL7WGT	450 G	15	S2	---	---	---	---
---	Post-shock end points 6SL7WGT	Low-frequency vibration	---	---	Ep	---	200	mVac
		Heater-cathode leakage	---	---	Ihk	---	45	µAdc
		Transconductance	---	---	Sm	1,040	---	µmhos
1301	Heater current 6SL7WGT		---	---	If	275	325	mA
	12SL7GT		---	---	If	138	162	mA
1211	Insulation of electrodes 6SL7WGT		4.0	S3	R	100	---	Meg
	12SL7GT		4.0	S3	R	10	---	Meg
	12SL7GT	See note 7	4.0	S3	R (a-a)	100	---	Meg

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 2 - Continued							
Ⓒ 1336	Heater-cathode leakage	See note 4	---	---	Ihk	---	30	µAdc
1256	Electrode current (2) (anode)	Ec = -5.5 Vdc (see note 6)	---	---	Ib	0	25	µAdc
1316	Amplification factor	See note 6	---	---	Mu	55	85	---
1308	Transconductance	See note 6	---	---	Sm	1,200	2,000	µmhos
Ⓒ 1101	Secureness of base, cap, or insert		---	---	---	---	---	---
Ⓒ 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A; Ec = -1.0 Vdc; Ehk = 100 V (see note 4)	---	---	---	---	---	---
---	Life-test end points (500 hours)	AC amplification or Transconductance	--- ---	--- ---	Ep Sm	6.8 1,040	--- ---	Vac µmhos

NOTES:

Ⓒ 1. Outline (9-11 EIA) except maximum overall length 3 inches (76.2 mm) and maximum seated height equals 2-7/16 inches (61.9 mm).

2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.

3. The mount shall exhibit no pronounced mechanical resonance below 100 Hz.

4. Test units connected in parallel.

5. This test to be performed at the conclusion of the holding period.

6. Test each unit separately.

7. Anode to anode resistance shall be measured with a potential of 500 Vdc.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MU
Navy -
Air Force - 11, 85
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2401-69)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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